MILITARY SPECIFICATION

# SWITCH, TRIGGER, SINGLE POLE, TWO STAGE, MOMENTARY

Inactive for new design after 6 April 1984.
For new design use MIL-S-8805/111.

This specification is approved for use by SA-ALC (82), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for a momentary two stage, single pole trigger switch to be used primarily in control stick grips.

## *2. APPLICABLE DOCUMENTS

*2.1 Government documents.

*2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| QQ-N-290 | Nickel Plating (Electrodeposited) |
| QQ-W-343 | Wire, Electrical And Nonelectrical, Copper, (Uninsulated) |
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |

MILITARY

| | |
|---|---|
| MIL-M-14 | Molding Plastics And Molded Plastic Parts Thermosetting |
| MIL-P-997 | Plastic-Material, Laminated, Thermosetting, Electrical-Insulating, Sheets, Glass Cloth, Silicone Resin |
| DOD-D-1000 | Drawings, Engineering, And Associated Lists |

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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| | | |
|---|---|---|
| | MIL-W-5086 | Wire, Electrical, 600-Volt, Copper, Aircraft |
| | MIL-C-5809 | Circuit Breakers, Trip-Free, Aircraft, General Specification For |
| | MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series General Specification For |
| | MIL-T-7928 | Terminals, Lug and Splice, Crimp Style, Copper |
| | MIL-P-15037 | Plastic-Material, Laminated, Thermosetting, Sheets, Glass-Cloth, Melamine-Resin |
| | MIL-E-17555 | Electronic And Electrical Equipment And Associated Repair Parts, Preparation For Delivery Of |
| | MIL-S-28786 | Switches, Preparation For Delivery Of |
| * | MIL-I-81023 | Inductor, 28V, D.C. Laboratory Test General Specification For |

STANDARDS

MILITARY

| | | |
|---|---|---|
| | MIL-STD-129 | Marking For Shipment And Storage |
| | MIL-STD-130 | Identification Marking Of U.S. Military Property |
| | MIL-STD-143 | Specification And Standards, Order Of Precedence |
| | MIL-STD-202 | Test Methods For Electronic And Electrical Components Parts |
| * | MIL-STD-810 | Environmental Test Methods |
| | MIL-STD-831 | Test Reports, Preparation Of |
| * | MIL-STD-889 | Dissimilar Metals |
| | MS25235 | Lamp, Incandescent, Single-Contact Bayonet |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

AIR FORCE

| | |
|---|---|
| 56C3002 | Grip Assembly, Controller, Aircraft, Type MC-2 |
| 56D3003 | Grip, External Shape, MC-2, Grip Assembly |

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)



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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

*3.1 First article. First article inspection shall be performed as specified in 4.3 and 6.4.

3.2 Components. The completed switch assembly shall consist of movable current-carrying conductors or blades, and actuating lever, and stationary contact members and terminals attached to an insulating base. The mechanism shall be secured to an insulating base with suitable mounting provisions for installation purposes.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Standard MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Use of commercial parts. Bolts, nuts, washers, lock rings, and similar fastening devices used for assembly may be selected from commercial sources, provided military or other standard parts are not specifically specified by this specification or associated standards. If such commercial parts are used, they shall possess suitable properties and shall be replaceable by military standard parts without alteration, and provided the corresponding military standard part numbers are referenced in the parts list, and, if practicable, on the contractor's drawings.

3.3.2 Standard parts. With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Material. Material shall be as specified herein. However, when a definite material is not specified, a suitable material shall be used. acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used.

3.4.2 Ferrous materials. Ferrous materials shall not be used for current-carrying parts.

*3.4.3 Plastic material. Except as specified herein, plastic material shall conform to MIL-M-14 for molded material, and MIL-P-997 or MIL-P-15037 for laminated material. Other types of plastic materials possessing superior characteristics may be used, provided the manufacturer submits acceptable evidence to the activity responsible for qualification that such material meets the performance requirements of MIL-M-14, MIL-P-997, or MIL-P-15037, whichever is applicable. In addition to these specifications, the plastic material shall not cause any hazard when operating under the extremes of environment and performance as specified herein. Such hazards may be defined as any tendency to support combustion, give off noxious gases in harmful quantities, give off any gases in a sealed housing that



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will cause contamination of the contacts or other parts of the component, or form current-carrying tracks when subjected to arcing conditions encountered when any of the tests in this specification are performed on the component.

*3.4.4 Metals. All metal parts (other than current-carrying parts) shall be of corrosion-resistant material or shall be cadmium or nickel plated to resist corrosion in accordance with type II of QQ-P-416 or QQ-N-290.

*3.4.4.1 Dissimilar metals. The use of dissimilar metals in direct contact with each other, which tend toward active electrolytic corrosion (particularly brass, copper, or steel in contact with aluminum alloy), shall not be acceptable. Metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces will be permitted. The use of dissimilar metals separated by suitable non-metallic insulating material will be permitted. Dissimilar metals are defined in MIL-STD-889.

3.5 Design and construction. The switch shall be of compact design and of sufficiently rugged construction to withstand the requirements of this specification. It shall be a normally open type incorporating a snap action which will assure good contact and rapid making and free breaking of the circuits. The construction shall be such as to assure free return of the movable contact member to the neutral "off" position when relieved of any external actuation forces.

3.5.1 Dimensions. The dimensions of the switch shall be such that it can be installed within the control stick grip of stick-controlled aircaraft. See figure 1.

3.5.2 Mounting. The mounting shall be of sufficiently rugged construction to withstand the mechanical stresses incident to its use on the control stick.

*3.5.2.1 Switch wiring. The switch wiring shall conform to the wiring shown on figure 1, using wire in accordance with QQ-W- 343 size AN-18.

3.5.3 Movable current-carrying conductors or blades. Movable current-carrying conductors or blades shall be made of brass, bronze, or silver alloy. When brass or bronze is used, it shall be heavily silver plated and shall have contact included thereon. Contacts shall be made of fine silver or an alloy thereof.

3.5.4 Actuating trigger. All portions of the actuating trigger subject to contact by operating personnel shall be made from molded plastic material in accordance with MIL-M-14. Metal portions, if used, shall be corrosion resistant, or shall be suitably treated to resist corrosion. The design shall be such as to preclude adjustment of trigger force or trigger travel to values other than those specified in 3.6 (d) and (e), respectively.

3.5.5 Insulating base. The insulating base may be constructed in either of the following ways:

(a) Separate base: A separate base shall consist of the most suitable insulating material acceptable to the procuring activity.



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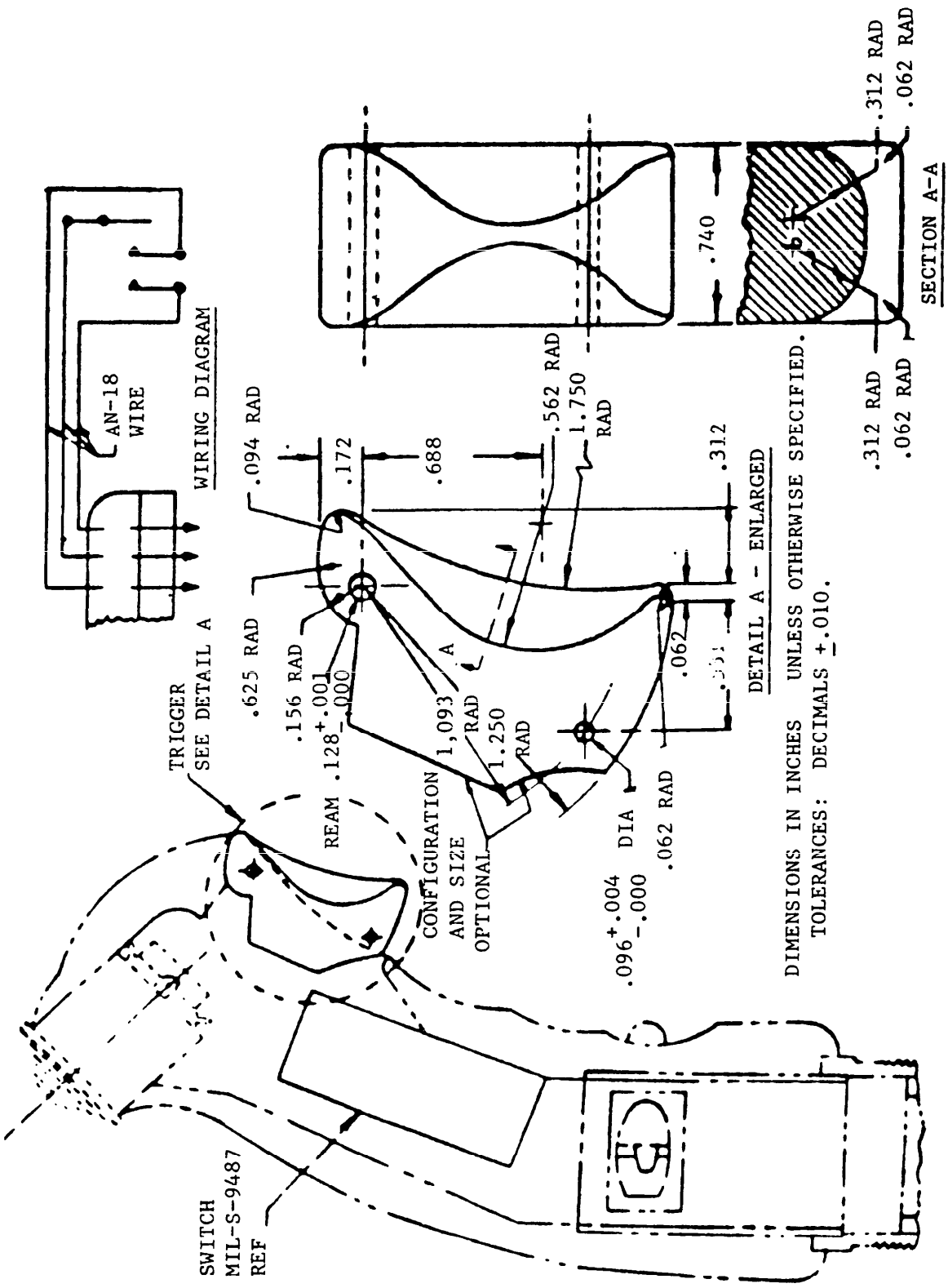



FIGURE 1. Grip assembly, controller aircraft.

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(b) Base integral with shell or cover: A base integral with shell or cover shall consist of a molded insulating material and cured under carefully controlled conditions of temperature, pressure and time. The finished part shall have an over-all smooth, high-finished surface; it shall be uniformly dense throughout its structure and shall be free from warpage, checks, cracks, blisters, projecting ridges, or chipped edges or surfaces.

*3.5.6 Shell or cover. The shell or cover shall be designed to completely enclose the contact elements of the switch. If made of steel, it shall be a positive lock joint and shall be plated in accordance with type II or III of QQ-P-416, or QQ-N-290, and of a class that is adequate to achieve the degree of protection required.

3.6 Performance. The switch shall conform to the following requirements when subjected to the applicable test specified in section 4:

(a) Operational characteristics in accordance with 4.7.2.

(b) Contacts and terminals: Stationary contact member and terminals shall be of sufficiently rugged construction to withstand the tests of 4.7.3. Terminals shall be of a size and configuration acceptable to the procuring activity. All terminals shall be hot-tin-dipped, or otherwise treated to facilitate soldering. Terminals, after plating, shall be given a protective coating or dip composed of 1 ounce of stearic acid (commercial) dissolved in 1 gallon of xylol (commercial). Terminals shall be dipped or coated in the compound at 20° to 23° C and shall then be dried. All threaded terminals shall have at least two full threads and shall be provided with a screw and lockwasher.

(c) Strength of lever pivot and stop: The switch shall be subjected to a 100-pound force applied at the tip of the trigger in accordance with 4.7.4.

(d) Trigger force: The force required to engage the first stage switch shall be 2-1/4 ± 1/2 pound. Between the first and second stage switches there shall be a detent action requiring 7-1/2 ± 1/2 pound trigger force to pass through, in order to provide a definite force indication to the pilot that the second switch is being actuated. After passing through the detent, the trigger force shall drop to 6 +0/-2 pounds, at which time both switches shall make contact.

(e) Trigger travel: Travel, measured at the bottom of the trigger shall be 7/32 ± 1/32 inch for each stage. The first-position contact shall not make with less than 1/8 inch movement. The detent shall be a distinct force which must be overcome. The second-position contact shall make between the detent and point of maximum travel and shall have a minimum of 1/32 inch overtravel.

(f) Dielectric withstanding voltage: Dielectric withstanding voltage shall be 1,000V root mean square (rms), plus twice rated voltage at sea level and 500V rms at 65,000 feet altitude when tested in accordance with 4.7.7.



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(g) Contact voltage drop: When tested in accordance with 4.7.8, the voltage drop shall not exceed 1 millivolt (mv).

(h) Short circuit: There shall be no welding or sticking of contacts, mechanical failure, or damage to the switch case, when tested as specifiied in 4.7.9.

(i) Mechanical endurance and temperature: There shall be no breakage or mechanical failure, when tested as specified in 4.7.10

(j) Electrical endurance: There shall be no evidence of electrical or mechanical failure, when tested as specified in 4.7.11.

(k) Overload: There shall be no mechanical failure and the switch shall be electrically operative on completion of the tests specified in 4.7.11.1.

(l) Resistive load: There shall be no mechanical failure and the switch shall be electrically operative on completion of the test specified in 4.7.11.2.

(m) Inductive load: There shall be no mechanical failure and the switch shall be electrically operative on completion of the test specified in 4.7.11.3.

(n) Current-carrying capacity: The switch shall make and break an inductive current of 5 amps, a resistive current of 10 amps, and lamp load current of 3 amps applied separately when tested as specified in 4.7.11.2, 4.7.11.3, and 4.7.11.4.

(o) Altitude: The switch shall operate satisfactorily at pressures ranging from 30 inches Hg down to 1.67 inches Hg (approximately 65,000 feet altitude) when tested as specified in 4.7.7 and 4.7.11.

(p) Lamp load: When tested as specified in 4.7.11.4, there shall be no mechanical failure, and the switch shall be electrically operative on completion of the test.

(q) Insulation: Molded or laminated material shall be used for insulation. The minimum creepage distance between any current-carrying parts and any other portions of the switch, other than insulating material, shall be 3/32 inch, when tested as specified in 4.7.7.

* (r) Vibration: The switch shall pass the vibration test Method 514.2 of MIL-STD-810, when tested as specified in 4.7.12.

(s) Shock: The switch shall pass the shock test Method 202 of MIL-STD-202, when tested as specified in 4.7.13.

(t) Acceleration: The switch shall pass the test for acceleration, when tested as specified in 4.7.14.

* (u) Salt spray: The switch shall pass the 96-hour salt spray test Method 101 of MIL-STD-202, when tested as specified in 4.7.15.



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* (v) Moisture resistance: The switch shall pass the test for moisture resistance Method 106 of MIL-STD-810, when tested as specified in 4.7.16.

* (w) Sand and dust: The switch shall pass the sand and dust test Method 510.1 of MIL-STD-810, when tested as specified in 4.7.17.

* (x) Explosion: The switch shall pass the explosion test Method 511.1 of MIL-STD-810, when tested as specified in 4.7.18.

*3.7 Interchangeability. All parts having the same manufacturer's part number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.8 Soldering: Where soldering is employed, only noncorrosive fluxes shall be use. Excess flux shall be removed. Solder shall not be used primarily for obtaining mechanical strength. Electrical connections shall be mechanically secure and electrically continuous before and after soldering.

*3.9 Screw threads. Screw threads shall be in accordance with MIL-S-7742. Threading of nonmetallic parts shall not be permitted.

3.10 Weight. The weight of the switch shall not exceed 0.20 pounds.

*3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The ON and OFF positions of the switch and the electrical terminals shall also be permanently marked for identification.

*3.12 Drawings. When the contract or order so provides, the following Air Force drawings may be obtained upon request to Engineering Division, San Antonio, ALC/MMED Kelly AFB, TX 78241.

| Drawing Number | Item Description |
|---|---|
| 56C3002 | Grip Assembly, Controller, Aircraft Type MC-2, Assembly Of |
| 56D3003 | Grip, External Shape, MC-2, Grip Assembly |

3.13 Workmanship. The switch shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirement as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirement.

4.2 Classification of inspections. Examination and testing of the switch shall be classified as follows:

* (a) First article inspection (4.3)
* (b) Quality conformance inspection (4.4)

*4.3 First article inspection. First article inspection shall consist of all the examinations and tests specified herein performed in the sequence specified in table I.

*4.3.1 Sampling instructions. Twelve switches shall be subjected to the First article inspection specified in 4.3.

*4.3.2 First article inspection report. A first article inspection report shall be prepared in accordance with MIL-STD-831. Three copies of the report shall be forwarded with the samples specified in 4.3.3.

*4.3.3 First article samples for the procuring activity. After the samples specified in 4.3.1 have been tested, samples shall be forwarded to the procuring activity as specified in the contract or order (see 6.2)."

*4.4 Quality conformance inspection shall consist of the following:

| | | |
|---|---|---|
| (a) | Individual tests | (4.4.1) |
| (b) | Sampling | (4.4.2) |

4.4.1 Individual tests. Each switch shall be subjected to the following tests as described under 4.7:

| | | |
|---|---|---|
| (a) | Examination of product | (4.7.1) |
| (b) | Operation | (4.7.2) |
| (c) | Insulation (1,000V rms at sea level) | (4.7.7.1) |

4.4.2 Sampling plans and tests.

4.4.2.1 Sampling plan A. One switch selected at random from each 100 or less produced on the contract or order shall be subjected to the following test as described under 4.7:

| | | |
|---|---|---|
| (a) | Individual tests | (4.4.1) |
| (b) | Terminal strength | (4.7.3) |
| (c) | Strength of lever pivot and stop | (4.7.4) |
| (d) | Trigger actuating force | (4.7.5) |
| (e) | Trigger actuating travel | (4.7.6) |
| (f) | Dielectric withstanding voltage | (4.7.7) |
| (g) | Voltage drop | (4.7.8) |
| (h) | Short circuit | (4.7.9) |
| (i) | Inductive load | (4.7.11.3) |

4.4.2.2 Sampling plan B. Unless otherwise specified, 12 switches selected at random from the first 25 items of the contract or order shall be subject to the following tests as specified under 4.7:



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TABLE I. Test sequence.

| No. | Tests | Paragraph | Sample Number | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 1 | Examination | 4.7.1 | 1(12) | 1(11) | 1(12) | 1(12) | 1(6)(12) | 1(6)(12) | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | Operation | 4.7.2 | 2(13) | 2(12) | 2(13) | 2(13) | 2(7)(13) | 2(7)(13) | 2(7) | 2(7) | 2(6) | 2(6) | 2(6) | 2(6) |
| 3 | Terminal strength | 4.7.3 | 3 | 3 | 3 | 3 | | | | | | | | |
| 4 | Strength of lever | 4.7.4 | 4 | 4 | 4 | 4 | | | | | | | | |
| 5 | Trigger force | 4.7.5 | 5 | 5 | 5 | 5 | 3(9) | 3(9) | 3(8) | 3 | 3 | 3 | 3 | 3 |
| 6 | Trigger travel | 4.7.6 | 6 | 6 | 6 | 6 | 4(10) | 4(10) | 4(9) | 4 | 4 | 4 | 4 | 4 |
| 7 | Withstanding voltage | 4.7.7 | 7(9)(11) | 7(9) | | | 11 | 11 | | | | | | |
| 8 | Voltage drop | 4.7.8 | | | 7 | 7 | | | | | | | | |
| 9 | Short circuit | 4.7.9 | | | 8 | 8 | | | | | | | | |
| 10 | Mech. endurance | 4.7.10 | | | | | | | 6 | | | | | |
| 11 | Overload | 4.7.11.1 | | | | | | | 5 | 5 | | | | |
| 12 | Resistive load | 4.7.11.2 | | | | | | | | | 5 | 5 | | |
| 13 | Inductive load | 4.7.11.3 | | | | | | | | | | | 5 | 5 |
| 14 | Lamp load | 4.7.11.4 | | | | | | | | 6 | | | | |
| 15 | Vibration | 4.7.12 | | | 9 | 9 | | | | | | | | |
| 16 | Shock | 4.7.13 | | | 10 | 10 | | | | | | | | |
| 17 | Acceleration | 4.7.14 | | | 11 | 11 | | | | | | | | |
| 18 | Salt spray | 4.7.15 | | | | | 8 | 8 | | | | | | |
| 19 | Moisture resistance | 4.7.16 | 8 | 8 | | | | | | | | | | |
| 20 | Sand and dust | 4.7.17 | 10 | 10 | | | | | | | | | | |
| 21 | Explosion | 4.7.18 | | | | | 5 | 5 | | | | | | |

NOTE: Numbers in ( ) indicate repeat tests to be conducted in specific sequence.





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| | | |
|---|---|---|
| (a) | Sampling plan A | (4.4.2.1) |
| (b) | Mechanical endurance | (4.7.10) |
| (c) | Electrical endurance | (4.7.11, except 4.7.11.3) |
| (d) | Vibration | (4.7.12) |
| (e) | Shock | (4.7.13) |
| (f) | Acceleration | (4.7.14) |
| (g) | Salt spray | (4.7.15) |
| (h) | Moisture resistance | (4.7.16) |
| (i) | Sand and dust | (4.7.17) |
| (j) | Explosion | (4.7.18) |

4.5 Rejection. Items failing to conform to all requirements of this specification shall be rejected.

4.6 Test conditions.

4.6.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.6.2 Test equipment and facilities.

4.6.3 Test fixture. Unless otherwise specified, the switch shall be tested in a test fixture so constructed as to simulate manual control stick grip operating conditions. The test fixture shall include associated circuit wiring for testing the switch with the current and voltage specified in table II.

TABLE II

Current and voltage values

| Resistive circuit | Inductive circuit | Lamp-load circuits |
|---|---|---|
| 10 amps at 28V dc<br>10 amps at 115V-400 cps | 5 amps at 28V dc<br>5 amps at 115V-400 cps | 3 amps at 28V dc<br>3 amps at 115V-400 cps |

4.7. Test method.

4.7.1 Examination of product. The switch shall be examined to determine conformance with this specification with respect to material, workmanship, design and construction, detail drawings, mounting provisions, and binding screws.

4.7.2 Operational characteristics. The switch shall be installed in the test fixture and each set of contacts tested separately in an electrical circuit having the rated current and voltage specified in table II. The switch shall be operated through its complete range and shall insure good electrical contact, positive current-carrying capacity, and rapid-making and free-breaking of each circuit. Any mechanical or electrical failure shall be cause for rejection.



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4.7.3 Terminal strength. A tensile load of 30 pounds shall be applied to each terminal, parallel to the plane of the terminal for a period of 1 minute. The switch shall be checked for short circuiting and shall operate satisfactorily both mechanically and electrically after this test. Any mechanical or electrical failure shall be cause for rejection.

4.7.4 Strength of lever pivot and stop. The switch shall be mounted in a control stick grip, in its normal position and subjected to a 100-pound force applied at the tip of the trigger, parallel to tip motion. Following this test there shall be no malfunction, damage, breakage, or short circuit, and the switch shall operate mechanically and electrically at a rated resistive load of 10 amps at 28V dc, for 10 cycles of operation.

4.7.5 Trigger actuating force. The switch shall be mounted in a control stick grip and subjected to forces applied to the tip of the trigger parallel with the trigger travel. The force required to engage the first stage switch shall be 2-1/4 ± 1/2 pounds. Then between the first and second stage switches there shall be evidence of a detent action requiring a force of 7-1/2 ± 1/2 pound to actuate the second stage switch. After passing through detent, the force shall drop to 6 +0,-2 pounds, and both switches will then be making contact. Failure of the above test shall be cause for rejection.

4.7.6 Trigger actuating travel. The switch shall be mounted in a control stick grip and the trigger actuating travel measured at the tip of the trigger shall be as follows. (Failure of this test shall be cause for will rejection.):

(a) Travel for each stage - 7/32 ± 1/32 inch.
(b) Travel to first contact position - 1/8-inch minimum.
(c) Overtravel after engaging second contact prior to engaging stop - 1/32-inch minimum.

4.7.7 Dielectric withstanding voltage.

4.7.7.1 At sea level. Switches shall be tested in accordance with Method 301 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Magnitude of test voltage: 1,000V plus twice rated voltage.
(b) Test voltage for acceptance inspection shall be maintained for at least 5 seconds.
(c) Nature of potential: Alternating current.
(d) Points of application of test voltage: Between mutually insulated parts and also between current-carrying parts and exposed or grounded metal parts.
    (1) Measurements shall be repeated for each trigger position.
(e) Momentary flashover or leakage current in excess of 10 micro-amperes as indicated by a suitable ammeter shall be considered a failure

*4.7.7.2 At reduced pressure. Switches shall be tested as specified in 4.7.7.1 and in accordance with Method 105 of MIL-STD-202. The following details and exceptions shall apply:

(a) Reduced pressure equivalent to 65,000 feet altitude.
(b) Dielectrical potential - 500V.

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4.7.8 Voltage drop. Switches shall be subjected to a potential of 2V to 4V dc applied through a suitable resistor so the current through the switches contacts shall be 0.1 amp. Three separate voltage-drop readings shall be taken across the switch terminals after each actuation of the switch for three successive actuations. The voltage-drop reading shall not exceed 1 mv.

*4.7.9 Short circuit. The switch shall be inserted in a circuit calibrated to a supply current equal to 60 times the rated resistives load of 10 amps at 28V dc. The switch shall be connected in series by 1-foot lengths of wire, to a thermal-type circuit breaker in accordance with figure 2. The wire size shall be chosen for use in free air as listed in MIL-W-5086. The terminals shall be in accordance with Specification MIL-T-7928. The circuit breaker in accordance with MIL-C-5809, will be of the same current rating as the resistive current rating of the switch and shall be inserted in the circuit. Calibration shall be made without the test switch, or switch leads in the circuit. With both the switch under test and the circuit breaker in a closed position, the circuit shall be closed manually by a third switch (SW1). A minimum of 2 minutes shall elapse between successive switch closings. The test shall be conducted 5 times. For double-throw switches, one half of the switches shall be tested in one position, and the remaining half shall be tested in the other position. Any welding or sticking of contacts, mechanical failure, or damage to the switch case shall be cause for rejection.

4.7.10 Mechanical endurance. The switch shall be subjected to 50,000 cycles of operation; 25,000 cycles at -65° ± 2° C and 25,000 cycles at 71° ± 2° C at a rate of 50 to 60 cpm. Following the test, the switch shall be examined. There shall be no deterioration, breakage, or malfunction of the trigger mechanism. Operational characteristics and trigger travel shall be as specified in 4.7.2 and 4.7.6, respectively.

4.7.11 Electrical endurance. Unless otherwise specified, the switches shall be subjected to a total of 50,000 make-and-break cycles in accordance with the conditions specified in 4.7.11.1 to 4.7.11.4, inclusive. The cycling rate shall be 15 ± 2 cpm. One half of the switches shall be tested with the load circuit connected to one set of contacts, and the remaining half of the switches shall be tested with the circuit connected to the other set of contacts. Any mechanical or electrical failure or damage shall be cause for rejection. Following the tests, the switches shall pass the operating characteristics specified in 4.7.2 and 4.7.8, except the voltage drop shall not exceed 1.5 mv.

4.7.11.1 Overload circuit test. Two switch shall be subjected to 50 make-and-break cycles of operation at a rate of 5 to 6 cpm while carrying 15 amps resistive current at 28V dc on both switch stages. The duty cycle shall be approximately 50 percent on and 50 percent off.

4.7.11.2 Resistive load.

(a) Direct-current load: Switch samples numbered 9 and 10 shall be subjected to 25,000 cycles of operation with a 10-amp resistive load at 28V dc. Test sample 9 shall have the load applied on the first stage contacts and shall be operated at sea level pressure in an ambient of 71° ± 2° C. Test sample 10 shall have the load applied on the second stage contact and shall be operated at a pressure altitude of 65,000 feet at room temperature. The duty cycle on both samples shall be approximately 50 percent on and 50 percent off.

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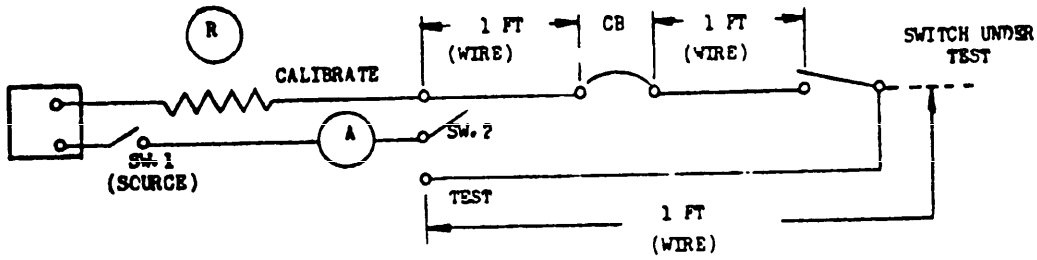



Figure 2. Circuit diagram for short-circuit test



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(b) Alternating-current load: Switch samples numbered 9 to 10 shall be tested as indicated in the foregoing paragraph except that the loads shall be 10 amp resistive at 115 V - 400 cycles ac.

4.7.11.3 Inductive load.

(a) Direct-current load: Switch samples 11 and 12 shall be subjected to 25,000 cycles of operation with a 5 amp inductive load at 28V dc obtained by means of MIL-I-81023 test inductors. Test sample 11 shall have the load applied on the first stage contacts and shall be operated at sea level pressure in an ambient temperature of 71° ± 2°C. Test sample 12 shall have the load applied on the second stage contacts and shall be operated at a pressure altitude of 65,000 feet at room ambient temperature. The duty cycle on both samples shall be approximately 25 percent on and 75 percent off.

(b) Alternating-current load: Switch samples numbered 11 amd 12 shall be tested as in the foregoing parapraph except that the loads shall be 5 amp inductive at 115 V - 400 cycle ac with a power factor of 0.70.

4.7.11.4 Lamp load.

(a) Direct-current load. Switch sample number 8 shall be subjected to 25,000 cycles of operation on the first stage contacts in a circuit having a steady-state lamp-load current of 3 amps at 28V dc. The lampload all be obtained by using (2) MS25235-311 lamps in parallel in a circuit in which the total resistance of the 28 V dc power supply plus the resistance of the leads shall not exceed 0.2 ohm. The lamps shall have a minimum cooling period of 15 seconds between successive operations. The duty cycle shall be approximately 30 percent on and 70 percent off.

(b) Alternating-current load: Switch sample number 8 shall be tested as in the foregoing paragraph except that the steady-state lamp load shall be 3 amps at 115V -400 cycle ac; obtained by use of tungsten lamps with a nominal rating not to exceed 200 watts at the voltage specified.

*4.7.12 Vibration. Two switches shall be subjected to the vibration test Method 514.2 of MIL-STD-810, except that the test shall be performed only in the 20° C to 35° C temperature range. There shall be no opening or closing of contacts as determined a by oscillographic or chronoscopic records. After this test, switches shall be subjected to and pass the test specified in 4.7.2.

*4.7.13 Shock. Two switches shall be tested in accordance with Method 213, Test condition A, MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting: The switch shall be secured in each of its three principal axes, in turn, to sufficient mass.

(b) Number of blows: Three separate shocks shall be applied in each direction of the three principal axes of the switch, (a total of 18 blows to be applied).



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(c) Desired deceleration: 50g units with a time duration of 11 ± 1 milliseconds.

There shall be no opening or closing of contacts as determined by an oscilloscope, oscillograph, or chronoscope. After this test, switches shall be subjected to and pass the test specified in 4.7.2.

4.7.14 Acceleration. Two switches mounted in a position most likely to cause malfunction, shall be subjected for a minimum of 1 minute to an acceleration force of 20g attained within 2 seconds. There shall be no opening or closing of contacts as determined by an oscilloscope, oscillograph or or chronoscope. After this test, switches shall be subjected to and pass the test specified in 4.7.2.

*4.7.15 Salt spray. Two switches shall be tested in accordance with Method 101 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition: A.

(b) Treatment after test: within 10 minutes after test, switches shall be washed for 5 minutes under running water not warmer than 37°C, accompanied by slight brushing, then dried for 6 hours in a force-draft oven at approximately 57°C.

(c) Following the drying period, switches shall be subjected to 50 cycles of operation of making and breaking a circuit having 10-amp resistive current at 28V dc.

There shall be no warping, cracking, or other damage, and the specified cycling shall be complete without failure.

4.7.16 Moisture resistance. Two switches shall be tested in accordance with Method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting: By normal mounting means on noncorrosive metal panel positioned 15 degrees from the vertical.

(b) Polarization: During steps 1 to 6, inclusive, a dc potential of 100V shall be applied between current-carrying parts and panel. Negative polarity shall be applied to panel. There shall be no electrical failures, and leakage current in excess of 1 milliampere will be considered a failure.

(c) Final measurements: Following a 24-hour period at 25° ± 5°C and 50 ± 5 percent relative humidity, operating characteristics shall be as specified in 4.7.2 and 4.7.8, except the voltage drop shall not exceed 1.5 mv. There shall be no breaking, cracking, spalling, or loosening of terminals.

*4.7.17 Sand and Dust. Two switches shall be subjected to the sand and dust test Method 510.1 of MIL-STD-810. During the test, switches shall be operated at 25° ± 2° C for 2,500 cycles and at 71° ± 2° C for 2.500 cycles of marking and breaking the tated inductive load of 28V dc, at a rate of 10 to 12 cpm. After this test, switches shall be subjected to and pass the test specified in 4.7.2 and 4.7.8, except the voltage drop shall not exceed 1.5 mv.



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*4.7.18 Explosion. Two switches shall be subjected to the explosion-proof test Method 511.1 of MIL-STD-810. When the switch is operated at its maximum rated dc inductive current, an explosion mixture in the test chamber shall not be ignited, whether or not an explosion occurs within the switch. After this test, the switches shall pass the tests specified in 4.7.2 and 4.7.8, except the voltage drop shall not exceed 1.5 mv.

4.8 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance with section 5.

5. PACKAGING

5.1 Packaging requirement. The requirements for packaging shall be in accordance with MIL-S-28786. Switches shall be preserved and packaged in accordance with level A or C of MIL-E-17555, as specified (see 6.2). They shall be packed in accordance with level A, B, or C, as specified (see 6.2), and marked in accordance with MIL-E-17555.

6. NOTES

*6.1 Intended use. The switch covered by this specification is intended for mounting in the control stick of fighter aircraft and serves as the electrical camera-gun control. Inactive for new design.

*6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable level of packing and packaging required (see 5.1).
c. Number of tested first article samples required, shipping, instructions for the samples and first article test report (see 4.3.2 and 4.3.3).
d. Whether sampling plan B tests are to be conducted (see 4.4.2.2).

6.3 Definitions.

a. Cycle: A cycle consists of mechanically moving the actuating lever through the two contact positions, with return to the original position accomplished solely by the internal snap-action mechanism of the switch.

b. Detent: Detent is sharp rise and fall in trigger force between the two switch positions. Purpose of the detent is to provide a definite force indication to the pilot when the second-stage switch is actuated.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of twelve units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the first article and documents.



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6.5 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-9487C | SWITCH, TRIGGER, SINGLE POLE, TWO STAGE, MOMENTARY |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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